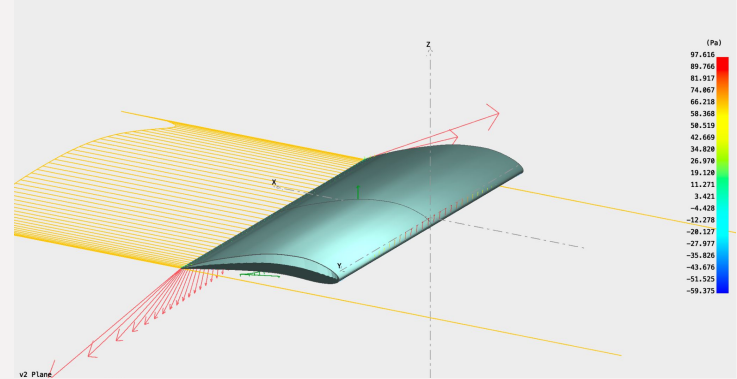
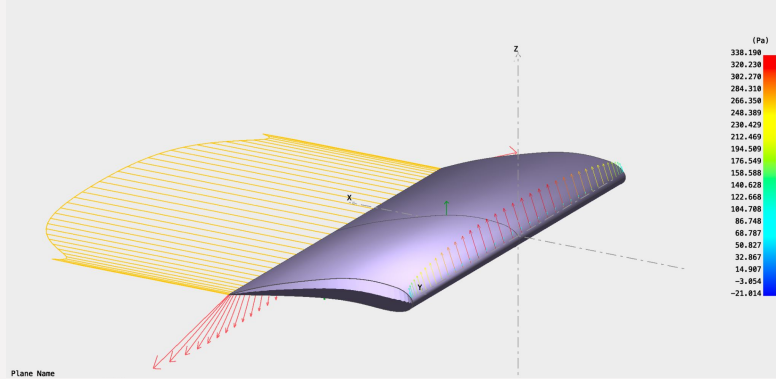


Prandtl of the Pack: FPC Part II

Presentation by Javi
Fernandez, Juan Perez, and
Devan Hernandez

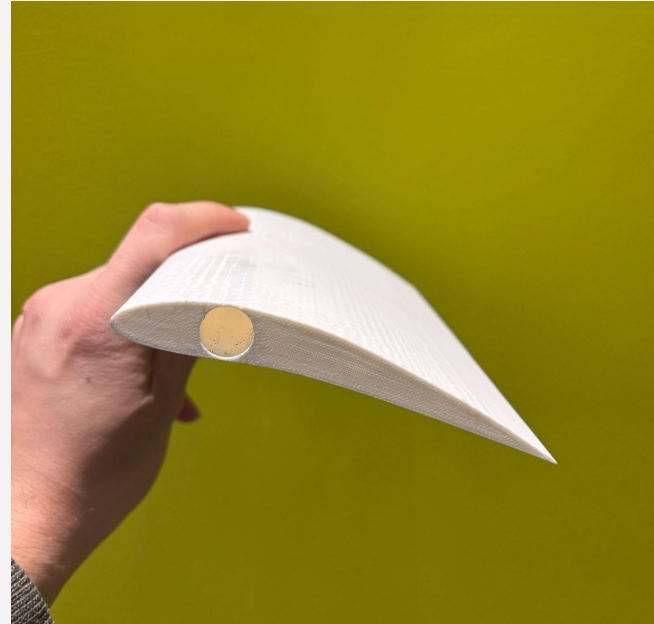
[Video Link](#)



Foil Designs



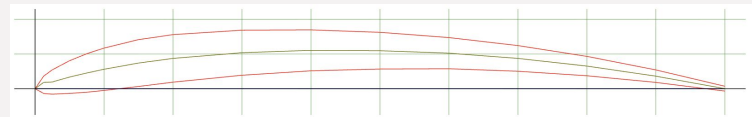
V1



V2

Design Process

NACA 2415 → Modified AH6407 → Final Design



First Prototype

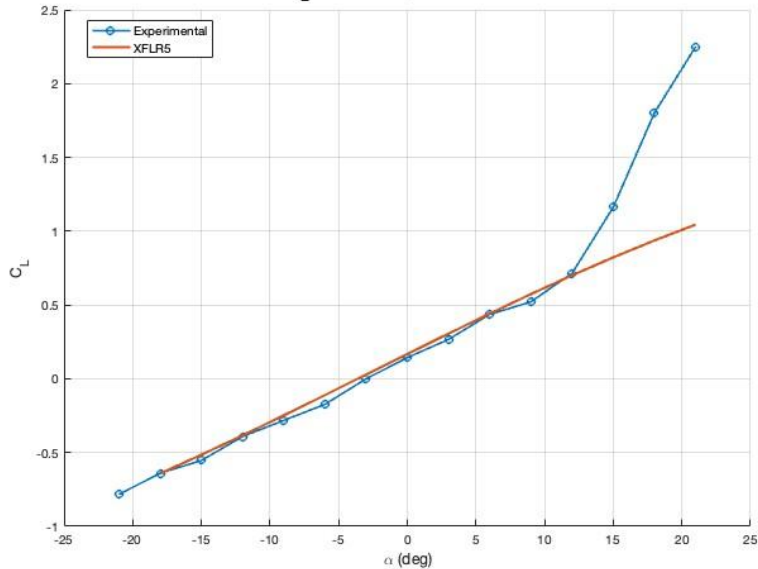
- Goal: Increase Aerodynamic Efficiency
 - Less Aggressive LE Radius
 - Increased Camber
 - Reduced Thickness

Final Prototype

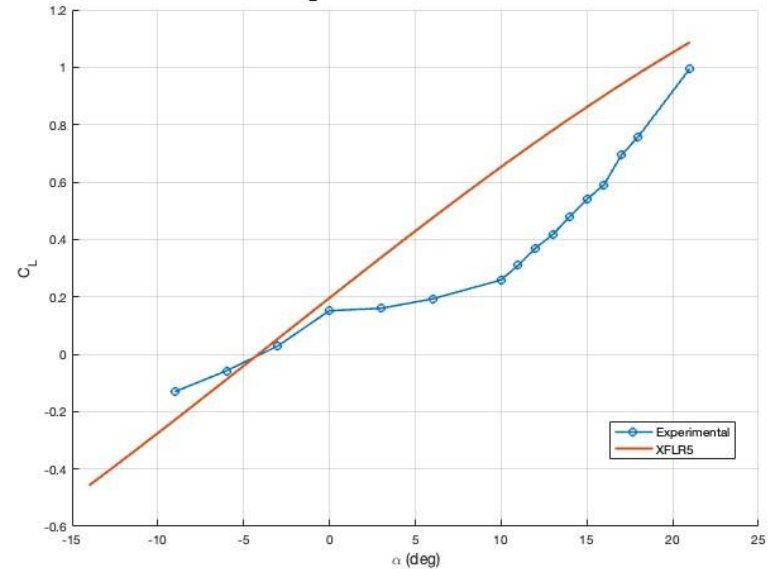
- Goal: Maximize Lift
 - Aggressive LE Radius
 - Slightly Increased Camber
 - Maintain 1st Design Thickness

Lift vs. Alpha Plots

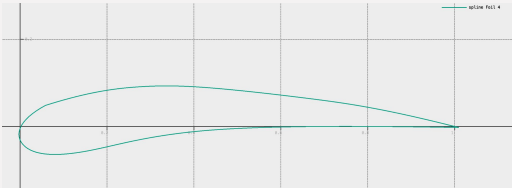
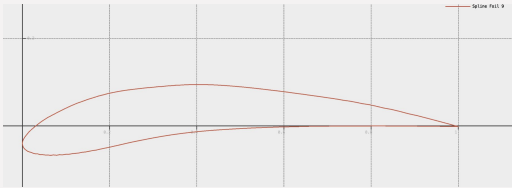
Foil 1: C_L vs α — Experimental vs XFLR5



Foil 2: C_L vs α — Experimental vs XFLR5



Final Metrics

	Version I	Version II
Profile		
Maximum Aerodynamic Efficiency	0.14 at $\alpha = 21^\circ$	0.57 at $\alpha = 3^\circ$
Maximum Lift at Max. L/D	2.31	0.178
Curvature at Max. L/D	N/A	$0.0479 \text{ (deg)}^{-2}$

Reflection

- Experimental behavior $\neq$ Simulated behavior
- Difficult to control for test conditions
- Main Sources of Error:
 - Reynold's Number Error
 - Imperfections in the print (grooves + sanding)
 - Incorrect Tare value
 - Experimental error relating to strain gauge angles
 - Lift Drag Crosstalk